

- 3 (a) The *lac* operon is a section of DNA present in the genome of the bacterium *Escherichia coli*. The structural genes of the operon are only fully expressed when *E. coli* is exposed to high lactose concentrations.

(i) Fig. 3.1 is a diagram showing the *lac* operon and a nearby region of the *E. coli* genome.

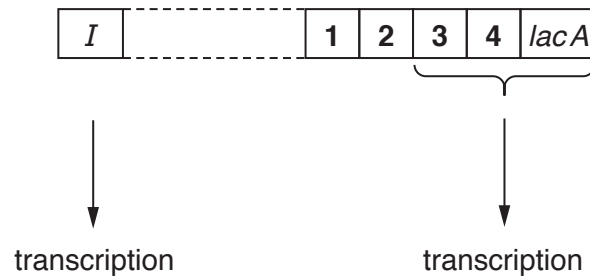


Fig. 3.1

Name sections **1**, **2**, **3** and **4** shown in Fig. 3.1.

- 1**
- 2**
- 3**
- 4**

[2]

- (ii) Certain enzymes need to be present inside *E. coli* for it to be able to take up and use lactose. When the genes of the *lac* operon are expressed, the enzymes β -galactosidase and lactose permease are produced in large quantities.

Outline the functions of β -galactosidase and lactose permease.

β -galactosidase

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.....

lactose permease

.....

.....

[2]

- (b) (i) Gene *I* is located a short distance away from the *lac* operon.

The product of gene *I*, a repressor protein, is a constitutive protein.

State what is meant by the term *constitutive protein*.

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.....[1]

- (ii) Explain why the structural genes of the *lac* operon are **not** expressed when lactose is absent.

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.....[3]

- (iii) A strain of *E. coli* has been produced with a mutation in gene *I*. Expression of this gene results in a non-functional repressor protein.

Suggest a negative effect that this mutation will have on this strain of *E. coli*.

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.....[1]

[Total: 9]